NeuroPredict AI

Precision Prediction. Dynamic Intervention.

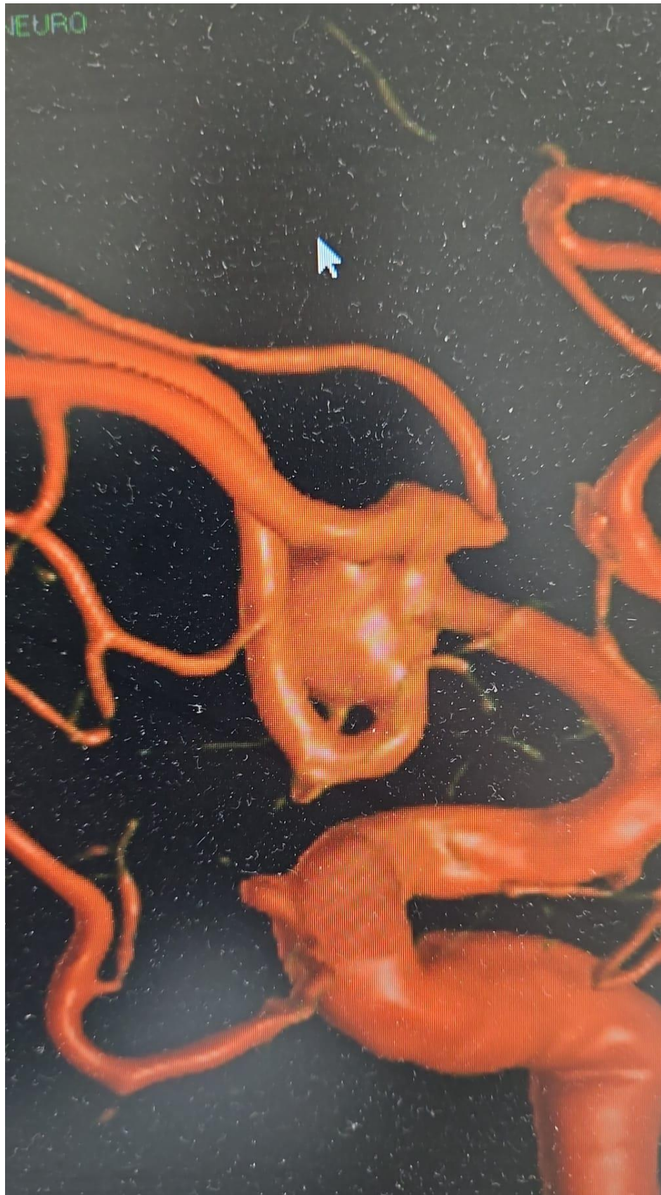
Problem : Treatment or surveillance? Treatment, How?

- Intracranial Aneurysms (IA) carry annual rupture risk
- Rupture is associated with high mortality and a substantial risk of severe long-term disability

Challenge #1: Predict which IA will rupture

- Aneurysms significantly vary in morphology, location, and anatomical complexity
- There are numerous treatment options, new devices are certified **every week**

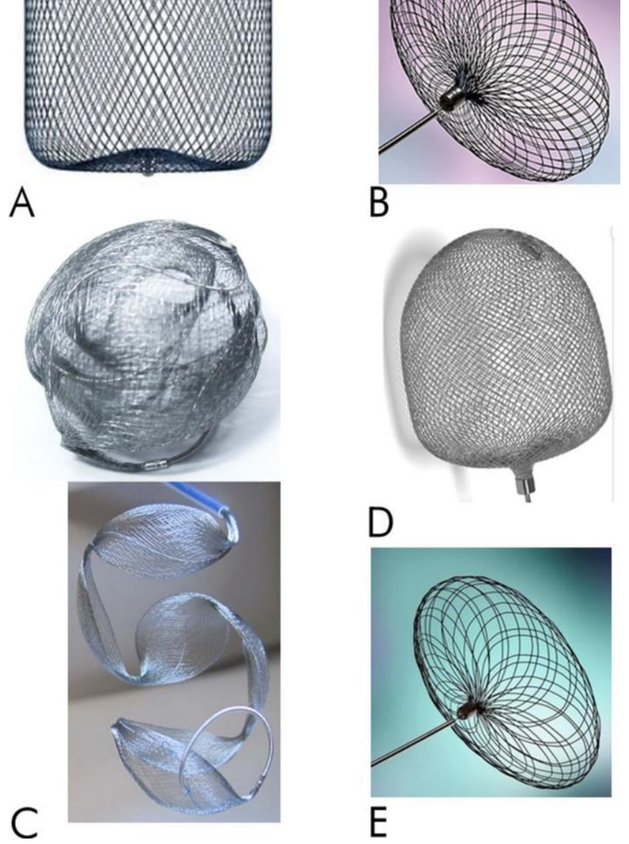
Challenge #2: Select the optimal therapeutic strategy

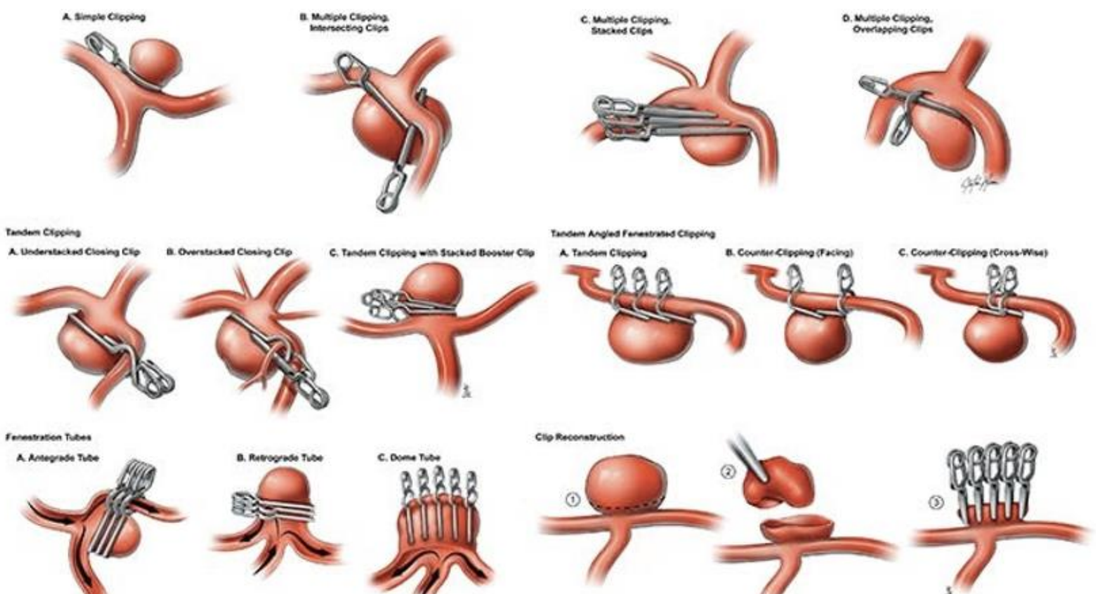


The Need

There is an unmet need for an intelligent imaging platform that can automatically:

- Detect and assess intracranial aneurysms on CTA scans
- Predict their risk of rupture
- Guide personalized treatment planning





Solution

Step 1: Automated Aneurysm Analysis

AI extracts 3D morphology and radiomic features directly from CTA/MRA

Fully automated, no manual segmentation.

Step 2: Rupture Risk Prediction

Integrate imaging features with clinical biomarkers

Deliver personalized rupture risk and support treatment decisions.

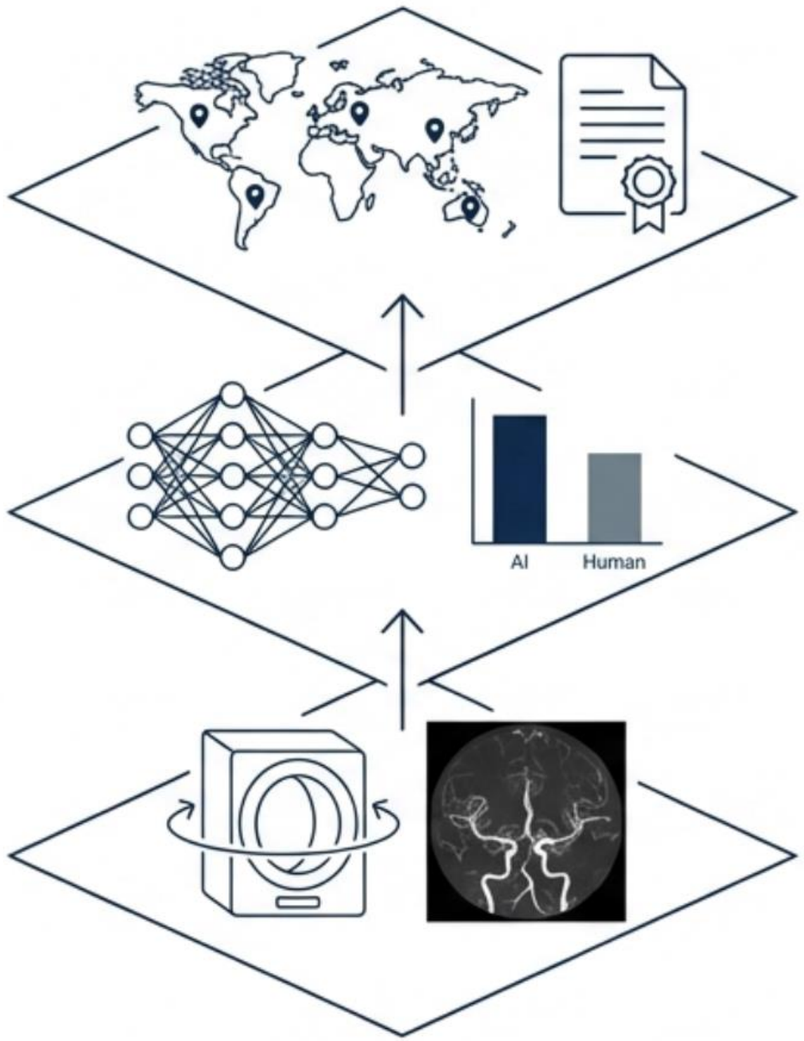
Step 3: Treatment Optimization

Simulates post-treatment blood flow

Identify optimal device configuration

Why Now?

- Imaging Quality
- Algorithmic Maturity
- Clinicians Receptivity
- Clinical Validation



The Market

- 6.5–7M Americans (~2–3%) - Living with an unruptured brain aneurysm
- ~70–80% are small (<10 mm) → ~4.5–5.5M US patients in surveillance / watch-and-wait
- ~30,000 ruptures annually in the U.S. - High mortality and severe morbidity
- Total lifetime cost of treatment Post rapture - $300,000–$1M+ (1, 2)
- Total lifetime cost of preventive intervention for unruptured intracranial aneurysms - $20,000–$80,000 (1, 2)

(1) Jaewoo Jayce Park et al., Interv Neuroradiol., 2025
(2) Stefan Strilciuc et al. , J Med Life,. 2021

Market Size

- TAM

~6,100 hospitals[1] (All U.S.; Annual license: $60K–$100K

Theoretical TAM: $366M–$610M

- SAM

~2,449 Certified stroke centers[2]; Annual license potential: $147M–$245M

- SOM

~300 high-acuity centers; Annual license potential: $18M–$30M

Realistic Early Adoption: 5–10% in Years 1–3

(1) https://www.aha.org/statistics/fast-facts-us-hospitals
(2) https://jamanetwork.com/journals/jamanetworkopen/fullarticle/2826106

Competition:

Category	Morphology	Clinical Factors	Hemodynamics (CFD)	Wall Biology	Automation	Treatment Optimization	Examples
PHASES	Basic (size/ location)	✓	✗	✗	✓	✗	PHASES score
UIATS	Basic	✓	✗	✗	✗	✗	UIATS framework
CFD Research Models	✓	✗	✓	✗	✗	✗	Academic CFD
Wall MRI Models	✓	✗	✗	✓	✗	✗	Fujimura studies
Stroke AI Platforms	Limited	✗	✗	✗	✓	✗	Viz.ai, Aidoc, Rapid AI
Device Vendor Tools	✓	✗	Limited	✗	Limited	✓ (device-specific)	Medtronic, Stryker, J&J
Our Platform	✓ Advanced 3D +	✓ Integrate	✓ Simulation-based	✓ Imaging-	✓ Fully Automated	✓ Personalized	—

Product:

https://ia-ai-score.base44.app/

Endovascular (EVT) | Neurosurgical (NT)

Patient Information

Patient ID / MRN

Enter patient identifier

Age (years) *

Enter age

Sex

Select sex

- [x] Smoking
- [x] Hypertension
- [ ] Dyslipidemia
- [ ] Cerebrovascular disease
- [x] Family history of SAH

Baseline mRS (0-6) *

Select mRS score

Aneurysm Characteristics

Location *

ICA cavernous

Size *

Select size

Morphology

Select morphology

Neck geometry

Select neck geometry

Neck surface

Select neck surface

- [x] Sac wall calcification
- [ ] Intraluminal thrombus
- [ ] Dissecting etiology
- [ ] Parent artery focal stenosis
- [ ] Collateral branch from sac
- [ ] Collateral branch from neck

EVT Planning

EVT Approach *

Select approach

Business Model

Target Customers

- Neurovascular and neurosurgery departments
- comprehensive stroke hospitals
- radiology departments.

Revenue Model

- Enterprise SaaS license: $60K-100K / per year (tiered by case volume)
- Per-case analysis fee: $300 per aneurysm evaluated

Pricing

- Pricing aligned with high economic burden of rupture ($300K–$1M per event)
- Supports billing under CPT 76376 / 76377 (3D rendering with interpretation)
- ROI driven by improved risk stratification and optimized treatment planning

Retention & Scalability

- Integrated into imaging and surgical workflow
- High switching cost → 5–7 year retention potential
- Scalable across high-volume neurovascular centers

Team

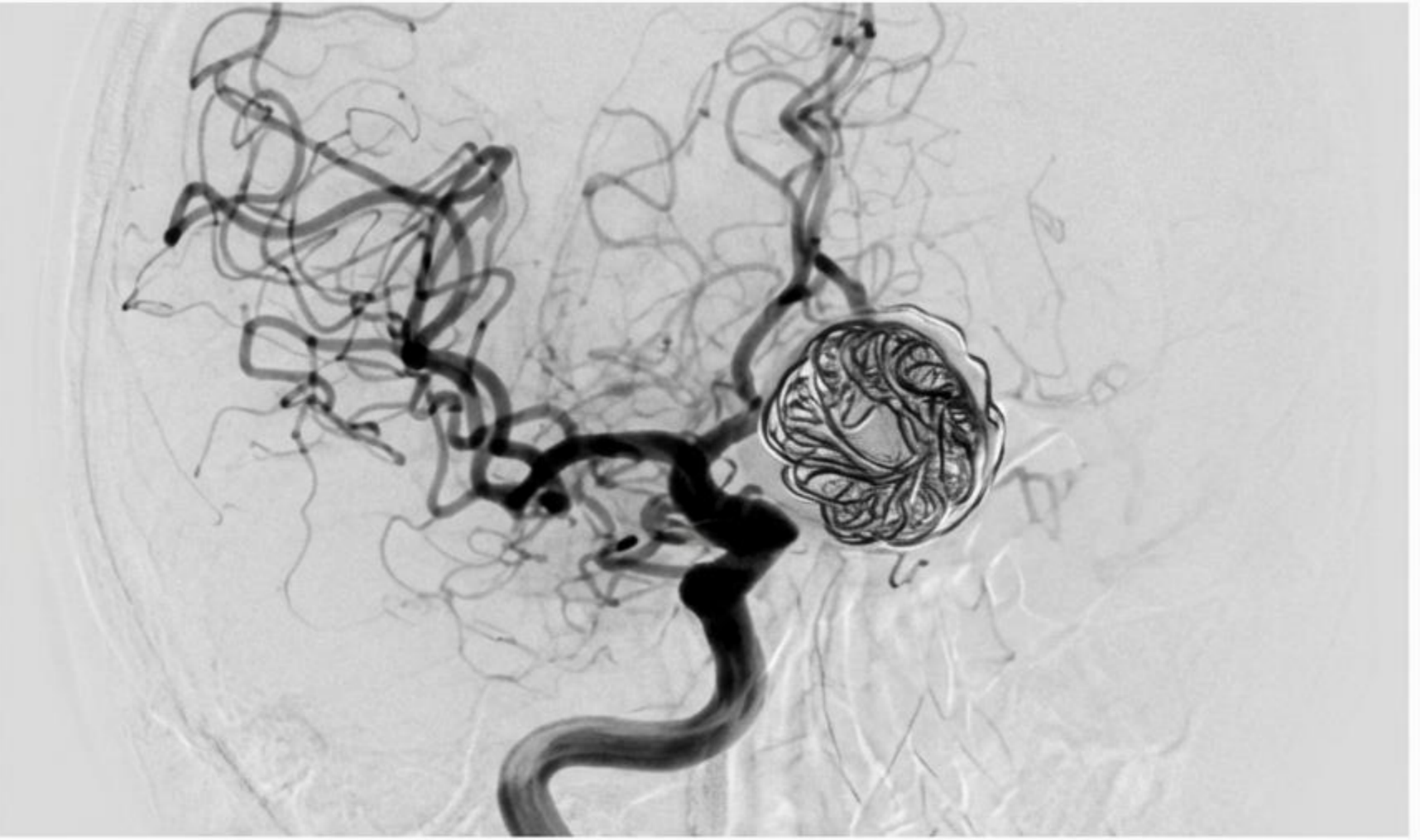
Johnny Kagan (CEO) - R&D manager with experience leading both HW and software development across the full product lifecycle.

Galit Ankri-Eliahoo, PhD (CTO) - Biomedical Engineer with vast experience leading multidisciplinary R&D and product initiatives.

Idan Levitan, M.D (CMO) - specialist neurosurgeon with a clinical focus on cerebrovascular disease, Medical 3D printing and SW development.

Ossnat Bar-Shira, PhD (CSO) - a neuroscience researcher specializing in data science and computational modeling.

Predicting the Unpredictable. Protecting the Patient.



We replace “wait and see” with “predict and prevent”.